



DOUGLAS COLLEGE
COMMERCE AND BUSINESS
ADMINISTRATION

SafeSight

A Full-Stack Predictive Workplace Safety Management

System with reference to

Fairmont Waterfront

VIDEO LINK:

https://drive.google.com/drive/folders/1LE58Eh13ZFkbEs37OQyys6fnp8NeUdGy?usp=drive_link



CSIS 4495 002: APPLIED RESEARCH PROJECT

MID TERM REPORT

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1. Introduction

Workplace health and safety is a critical domain within Human Resource Management, particularly in high-risk service industries such as hospitality. Hotels operate complex environments where employees are exposed to physical, operational, and ergonomic hazards across departments including Housekeeping, Food and Beverage, Engineering, and Front Office. Effective management of workplace safety is essential to protect employees, ensure regulatory compliance, and maintain operational efficiency (Cooper, 2000).

In British Columbia, organizations are required to comply with occupational health and safety regulations enforced by WorkSafeBC, which mandate accurate incident reporting, investigation, and documentation (WorkSafe BC, 2025). Despite these requirements, many organizations continue to rely on manual or fragmented systems for managing safety data, limiting visibility into incident trends and reducing the ability to proactively address workplace risks.

This research is positioned within the domain of digital workplace safety management systems, combining HR information systems, data visualization, and predictive analytics. SafeSight is a proposed full-stack workplace safety management platform designed specifically for the operational environment of the Fairmont Waterfront. The system aims to support HR and departmental managers by digitizing incident reporting, improving compliance with WorkSafeBC requirements, and providing real-time visibility into workplace safety trends. By combining role-based portals, data visualizations, and predictive analytics, SafeSight enables proactive identification of safety risks and supports informed decision-making within the hotel's hospitality operations.

1.1 Overview of the Background

Organizations across various industries are increasingly adopting digital systems to manage workplace operations, including health and safety management. Traditionally, workplace safety incidents have been recorded using paper-based forms or basic spreadsheet systems, which often result in delayed reporting, inconsistent data, and limited analytical capability. These limitations reduce an organization's ability to identify patterns, assess risks, and take preventive action.

In the hospitality industry, the complexity of operations and the diversity of job roles further increase safety management challenges. Employees frequently work under time pressure, handle

equipment, and perform repetitive physical tasks, making them more vulnerable to workplace injuries. As a result, hotels must manage large volumes of safety-related data while ensuring timely communication between HR teams, managers, and regulatory bodies.

Advancements in web technologies, data visualization, and predictive analytics provide new opportunities to modernize workplace safety management (Stone, et al., 2015)

. Full-stack web applications enable centralized data collection, role-based access through HR and manager portals, and real-time visualization of safety performance. Additionally, predictive analytics allows organizations to move beyond reactive reporting by forecasting potential risks and supporting proactive safety interventions.

This research builds on these technological advancements by proposing a full-stack, role-based workplace safety system that integrates incident management, regulatory reporting, and predictive insights within a single platform.

1.2 Research Context

This research is situated within the growing need for organizations to adopt digital solutions that improve workplace health and safety management while ensuring regulatory compliance. In British Columbia, employers are legally required to follow occupational health and safety standards established by WorkSafeBC, which include timely incident reporting, investigation documentation, and continuous monitoring of workplace risks.

In practice, many organizations particularly in the hospitality industry continue to manage safety incidents using manual processes or fragmented digital tools. These approaches limit data integration, delay reporting, and provide minimal analytical insight for HR professionals and managers. As a result, safety management is often reactive, focusing on incidents after they occur rather than preventing them.

At the same time, advancements in web technologies, data visualization, and predictive analytics have created opportunities to modernize safety management systems. Role-based web applications can support distinct user needs through HR portals and Manager portals, while centralized data storage enables real-time dashboards and automated reporting aligned with regulatory requirements.

Within this context, this research proposes the development of SafeSight, a full-stack, web-based workplace safety platform designed to support incident management, visual analytics, predictive risk assessment, and WorkSafeBC reporting. The system is developed as an applied case study within a hospitality environment, demonstrating how integrated digital solutions can enhance proactive safety management and informed decision-making.

1.3 Problem Statement

In high-pressure luxury hospitality environments like the Fairmont Waterfront, workplace safety management is frequently undermined by a reliance on manual, fragmented, or disconnected systems. These legacy processes create critical "data silos" between frontline staff, department managers, and HR teams, resulting in significant communication latency and a lack of centralized visibility into incident trends. This infrastructure deficit hinders the immediate identification of hazards and delays the production of mandatory WorkSafeBC compliance documentation, often leading to administrative burnout and potential regulatory penalties. Without integrated, role-based digital tools or predictive insights, safety management remains purely reactive, leaving the organization vulnerable to recurring injuries and operational inefficiencies. To address these gaps, SafeSight provides a unified full-stack platform designed to automate reporting, visualize risk trends, and transition the hotel's safety posture to proactive, data-driven prevention.

Research Questions

1. How can a full-stack, role-based safety system improve the speed and accuracy of incident reporting and monitoring?
2. In what ways do interactive dashboards and data visualizations support managerial safety decision-making?
3. To what extent can predictive analytics identify future workplace safety risks based on historical patterns?
4. How can automated reporting modules streamline compliance with WorkSafeBC statutory requirements?

2. Literature Review

Workplace health and safety management has increasingly shifted from manual record-keeping to digital systems that support incident reporting, monitoring, and regulatory compliance (Shah, 2026). Previous studies highlight that effective safety management systems play a critical role in reducing workplace injuries, improving compliance, and supporting proactive risk prevention. With advancements in web technologies, data visualization, and predictive analytics, organizations now have opportunities to enhance traditional safety practices through integrated digital platforms.

Hoteliers normally follow local regulations when designing a hotel's safety and security systems, but which system features hotel internal staff perceive to be the most important and which they expect to find in place remain open questions (Chan & Lam, 2013).

The Evolution of OHSMS in Global and Canadian Contexts

The concept of an Occupational Health and Safety Management System (OHSMS) has matured significantly since its inception in the late 1990s. Early frameworks, such as the OHSAS 18001 and various regional guidelines (British Standard Institution, 1996; ILO, 2001; Gallagher, et al., 2003), established the baseline for managing physical hazards. However, the transition to ISO 45001:2018 marked a paradigm shift, moving safety from a "siloed" administrative task to a core leadership responsibility.

Contextualizing OHSMS for the Canadian Hospitality Sector

For a landmark Canadian institution like the Fairmont Waterfront, the application of OHSMS is no longer just about meeting WorkSafeBC regulations; it is about organizational resilience. In the current post-pandemic landscape (2020–2026), the definition of "occupational health" has expanded to include,

- **Psychological Safety & Mental Wellbeing:** Research indicates that 87% of Canadian hospitality workers have experienced burnout post-pandemic (Future Skills Centre, 2024). Modern OHSMS must now integrate mental health resources to address the high-pressure nature of luxury service.

- **Musculoskeletal Injury (MSI) Prevention:** As high-end hotels return to peak occupancy, the ergonomic risks for housekeeping and guest services specifically regarding repetitive tasks remain a primary focus of Canadian safety audits (WorkSafeBC, 2025).
- **Integration with HRM:** Recent literature suggests that OHSMS is most effective when fused with Human Resource Management (HRM). This "total worker health" approach aligns safety performance with employee retention, which is critical given the persistent labor shortages in the Vancouver hospitality sector (The HUB International, 2026).

2.1 Identification of Research Gaps and the SafeSight Rationale

While the theoretical frameworks for OHSMS are well-established, a critical review of current safety practices in high-end hospitality specifically within the Canadian regulatory context of WorkSafeBC reveals four primary limitations. These gaps form the empirical justification for the development of the SafeSight full-stack system.

The Temporal Gap: Latency in Incident Response

Existing safety management often relies on fragmented communication channels (e.g., paper logs, emails, or verbal reports). In a high-traffic environment like the Fairmont Waterfront, this creates a "latency gap" where hazards remain active between the time of discovery and the time of management notification. Research suggests that "stale data" is a leading contributor to secondary accidents (Nelson, et al., 2013). SafeSight aims to address this by implementing a real-time full-stack architecture that synchronizes field reports with management dashboards instantaneously.

The Integration Gap: Stakeholder Silos

A significant limitation in current hotel operations is the "siloing" of safety data. HR departments, Department Managers (e.g., Housekeeping, F&B), and Safety Officers often utilize disparate systems to track the same incident. This lack of integration leads to data duplication and inconsistent reporting. SafeSight addresses this by providing a unified "Single Source of Truth," allowing multi-level access to a centralized database that serves the specific needs of each stakeholder from HR compliance to operational hazard mitigation.

The Regulatory Friction Gap: WorkSafeBC Compliance

According to Workers Compensation Act (2026) and WorkSafeBC (2025) Compliance with the Workers Compensation Act requires rigorous documentation and strict filing deadlines (e.g., Form 7 or Preliminary Investigation Reports). The administrative burden of manually transcribing incident data into regulatory formats often leads to "compliance fatigue," where accuracy is sacrificed for speed. SafeSight introduces an automated reporting module designed to map internal incident data directly onto WorkSafeBC-compliant templates, thereby reducing administrative overhead and ensuring 100% filing accuracy.

The Analytical Gap: Descriptive vs. Prescriptive Output

Current OHSMS implementations are largely descriptive they record what happened in the past but offer little insight into how to prevent future occurrences (Lindholm, et al., 2024). There is a lack of localized systems that provide active recommendations. SafeSight seeks to bridge this gap by transitioning from a passive record-keeping tool to a prescriptive analytics platform. By analyzing historical incident trends within specific Fairmont departments, the system generates actionable safety recommendations and heatmaps to guide preventative maintenance and training.

2.2 Research Hypotheses

This research aims to verify the following testable predictions through the development of SafeSight.

- Hypothesis 1 (H1): The implementation of a full-stack, real-time reporting system will reduce the time between incident occurrence and management notification by at least 70% compared to manual/paper-based methods.
- Hypothesis 2 (H2): Automated report generation will significantly decrease the rate of human error and missing data fields in WorkSafeBC documentation.
- Hypothesis 3 (H3): The use of heatmaps and predictive analytics will enable department managers to identify high-risk areas (e.g., the kitchen or loading docks) before recurring incidents occur, leading to a measurable decrease in "Lost Time" accidents.

2.3 Research Assumptions

For this study to be valid, the following conditions are assumed,

- **Stakeholder Participation:** It is assumed that staff and managers at the Fairmont Waterfront will provide honest and timely data when using the SafeSight interface.
- **Data Consistency:** It is assumed that historical incident data provided by the hotel for the predictive module is accurate and representative of typical operations.
- **Infrastructure Reliability:** It is assumed that the hotel's existing internal network can support a web-based, full-stack application without significant downtime.
- **Technical Literacy:** It is assumed that the target users (HR, Managers, and Staff) possess the basic digital literacy required to navigate a role-based dashboard.

2.4 Potential Benefits of SafeSight

The successful development of SafeSight offers significant advantages to the Fairmont Waterfront.

For HR and Senior Management

- **Unified Oversight:** A single dashboard providing a "bird's-eye view" of safety across all departments.
- **Reduced Liability:** Faster, more accurate WorkSafeBC filing reduces the risk of fines and increases the hotel's standing during external audits.

For Department Managers (F&B, Housekeeping, Maintenance)

- **Reduced Admin Burden:** Automating the "paperwork" allows managers to spend more time on guest experience and staff training.
- **Data-Driven Decisions:** Instead of guessing where the risks are, managers can use SafeSight heatmaps to allocate resources or safety signage to specific "hotspots."

For the Fairmont Waterfront Organization

- **Brand Excellence:** Reinforces Fairmont's reputation as a "Top Employer" in Canada by demonstrating a cutting-edge commitment to employee wellbeing.
- **Financial Savings:** By predicting and preventing injuries, the hotel reduces costs associated with worker compensation claims and temporary staffing.

3. Summary of the Initially Proposed Research Project

The proposed project, SafeSight, is a specialized full-stack web application designed to modernize occupational health and safety management within the luxury hospitality sector, specifically tailored for the Fairmont Waterfront. The project addresses the inefficiencies of manual safety reporting by integrating real-time data logging, automated regulatory compliance, and predictive risk modeling.

3.1 Core Objectives

The primary goal is to transition from reactive to proactive safety management through five key pillars:

- ***System Development***

Creating a role-based (Managers, HR) interface for real-time incident logging.

- ***Visualization***

Using interactive heatmaps and dashboards to identify high-risk hotel zones (e.g., kitchens vs. lobby).

- ***Predictive Modeling***

Utilizing machine learning to forecast potential safety "hotspots" based on historical data.

- ***Compliance Automation***

Reducing administrative burden by at least 50% through automated WorkSafeBC report generation.

- ***Evaluation***

Validating the system's efficiency and user ease-of-use through simulated deployment.

3.2 Technical Framework & Methodology

The project follows an Evolutionary Prototyping Model within an Applied Design Research (ADR) framework. The development is structured into a four-phase System Development Life Cycle (SDLC):

1. **Phase 1 (Analysis):** Gap analysis of current manual logs and mapping WorkSafeBC regulatory requirements.

2. **Phase 2 (Development):** Building the three-tier architecture (Node.js/Express backend, MySQL database, and JavaScript frontend).
3. **Phase 3 (Intelligence):** Integrating a Python/Flask prediction module using scikit-learn to analyze 478 industry-specific injury records.
4. **Phase 4 (Validation):** Quantitative benchmarking of reporting speed and Qualitative User Acceptance Testing (UAT).

3.3 Technology Stack

SafeSight leverages a modern, scalable stack to ensure accessibility across hotel departments:

- **Frontend:** HTML, CSS, JavaScript, and **Chart.js** for data visualization.
- **Backend:** **Node.js** and **Express.js** managing REST APIs.
- **Database:** **MySQL** for relational data storage of incident logs and corrective actions.
- **AI/Analytics:** **Python (Flask)** and **scikit-learn** for regression analysis and risk forecasting.

4. Expected Impact

The project aims to deliver a robust technological artifact that not only streamlines WorkSafeBC compliance but also provides "Safety Intelligence." By visualizing trends and predicting risks, SafeSight enables Fairmont leadership to make data-driven decisions, ultimately reducing workplace injuries in high-frequency departments like Housekeeping and Food & Beverage.

4. Changes to the proposal

4.1 Justification for Changes

1. Shift to React.js for Frontend Development

- *The Change:* Moving from vanilla JavaScript to React.js for the presentation layer.
- *Justification:* While vanilla JS is functional, the Managerial Dashboards and Heatmaps require complex state management (e.g., filtering incidents by department or date without reloading the page).
- *Research & Decision:* Comparisons showed that React's virtual DOM provides faster updates for real-time "hotspot" visualizations. This ensures the UI remains responsive even when handling 478+ historical records.

2. Enhanced Predictive Modeling (Random Forest vs. Linear Regression)

- *The Change:* Upgrading the prediction module from simple regression to Random Forest Classification.
- *Justification:* Initial assumptions that incident frequency followed a linear path were proved incorrect by the OSHA 721110 data. Hotel injuries (burns vs. slips) are categorical and non-linear.
- *Research & Decision:* Random Forest handles categorical variables (e.g., "Kitchen," "Lobby," "Laundry") more accurately, providing better "Safety Intelligence" for HR personnel.

3. Implementation of a Decoupled API Architecture

- *The Change:* Separating the Node.js business logic from the Python (Flask) analytics engine.
- *Justification:* Mixing Python (for AI) and Node.js (for user management) in a single environment creates dependency conflicts.
- *Research & Decision:* By using a RESTful bridge, the system becomes more stable. Node.js handles the high-concurrency reporting, while the Python Flask service is called only when a manager requests a "Predictive Risk Report."

4. Localized Mapping of OSHA Data

- *The Change:* Moving from using raw OSHA data to Geographically Mapped Synthetic Data.
- *Justification:* Raw OSHA data lacks the physical context of the Fairmont Waterfront. To make the Heatmaps (Objective 2) valuable, the data had to be "mapped" to specific hotel coordinates (e.g., Floor 3, East Wing Kitchen).
- *Research & Decision:* This allows for a more realistic User Acceptance Testing (UAT) phase, as managers can see "hotspots" in areas they recognize rather than abstract data points.

5. Refined Compliance Timeline

- *The Change:* Extending the Phase 4 (Validation) timeline by two weeks.
- *Justification:* Initial estimates underestimated the complexity of mapping 30+ data fields to the specific WorkSafeBC Form 7 PDF structure.
- *Research & Decision:* To ensure the "50% reduction in admin time" (Objective 4) is met, more time was allocated to "Field Mapping" to ensure the auto-generated PDF is 100% legally compliant.

5. Project Planning and Timeline

5.1 Project Responsibilities

As the sole researcher and developer for SafeSight, the role is divided into three primary "hats" that you will wear throughout the term:

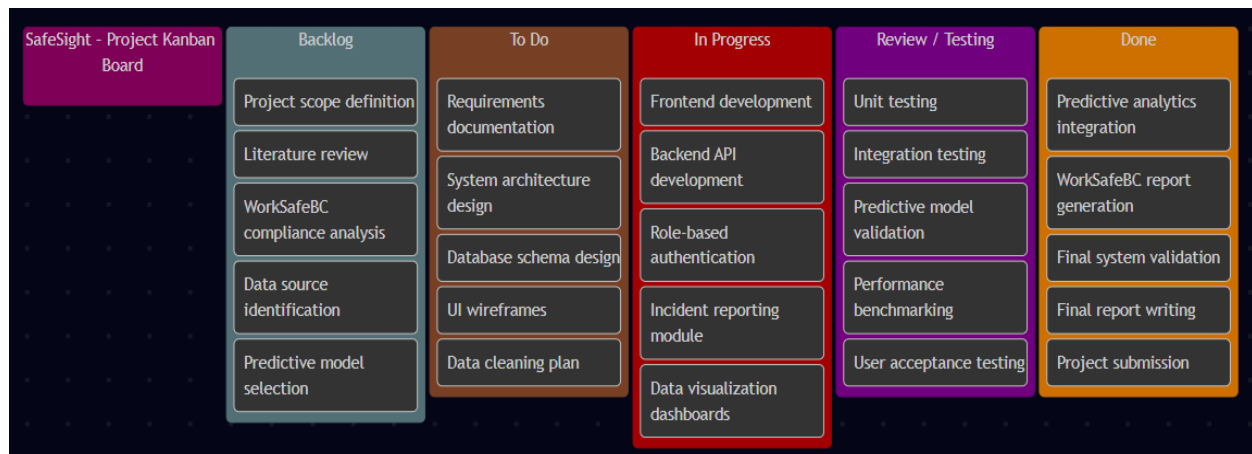
- **System Architect & Full-Stack Developer:** Responsible for the end-to-end technical build, including the React.js frontend, the Node.js API, and the MySQL database design.
- **Data Scientist:** Handle the cleaning of the OSHA 721110 dataset and the development of the Python/Flask predictive module using scikit-learn.
- **Compliance & QA Lead:** Manage the mapping of WorkSafeBC regulatory requirements and conduct all User Acceptance Testing (UAT) to validate the system's efficiency.

Updated Milestones

- **Milestone 1 (Data Foundation):** Successful connection between the React frontend and MySQL database.
- **Milestone 2 (Intelligence Bridge):** Establishing a secure REST API connection between your Node.js server and the Python/Flask prediction engine.
- **Milestone 3 (The "Efficiency" Test):** Successfully generating a WorkSafeBC Form 7 PDF with a single click, populated with real database values.

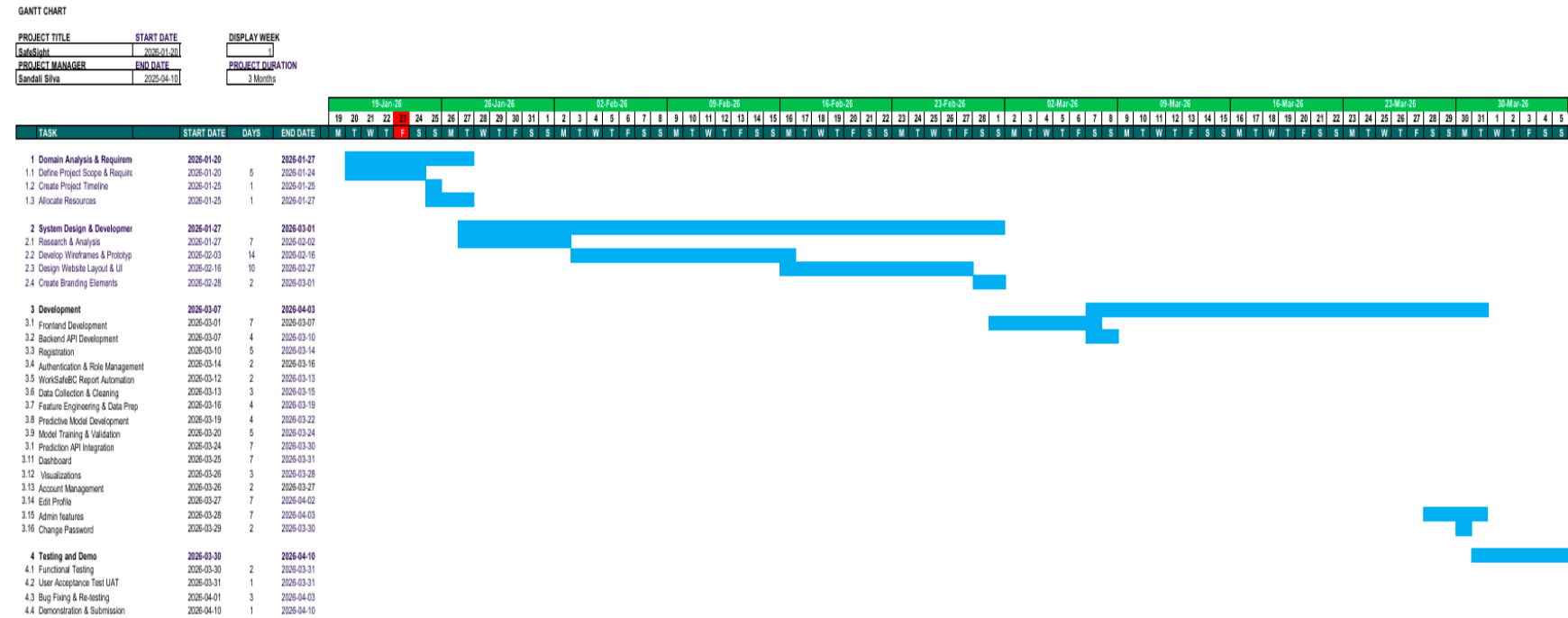
To ensure structured progress and timely completion, this project follows a defined timeline supported by both a Gantt chart and a Kanban board, outlining key phases, milestones, deadlines, and deliverables.

5.1 KANBAN



5.2 Gantt Chart

The following chart visualizes the overlap between the Predictive Modeling and Compliance Mapping, ensuring the backend is ready before the final validation begins.



5.3 Research Project Timeline and Milestones (SafeSight)

Phase	Duration (Weeks)	Key Activities	Milestones	Deliverables
Phase 1: Domain Analysis & Requirements Gathering	3–5	- Gap analysis of simulated manual incident logs - Stakeholder mapping (HR, Managers) - WorkSafeBC Form 7 and regulatory requirements analysis	- Gap analysis completed - Roles and permissions defined - Compliance mapping finalised	- Requirements Specification Document - Stakeholder & Role Matrix - Regulatory Alignment Document
Phase 2: Full-Stack System Architecture & Development	6–8	- Three-tier system architecture design - Frontend development (incident reporting & dashboards) - Backend API and authentication setup - MySQL database design and implementation	- Frontend prototype completed - Backend APIs operational - Database schema implemented	- System Architecture Diagram - Functional SafeSight Core Platform - Database Design Documentation
Phase 3: Predictive Analytics & Data Visualization	9–11	- Data cleaning and standardisation - Predictive modelling using regression and pattern analysis	- Clean dataset ready - Prediction model implemented	- Predictive Analytics Module - Data Visualization Dashboards - Model Validation Report

		• REST API integration for predictions • Dashboard visualizations (heatmaps, trends)	• Dashboards fully functional	
Phase 4: Testing, Evaluation & Validation	12–14	• Unit and integration testing • Performance benchmarking vs manual reporting • User Acceptance Testing (UAT)	• System testing completed • Performance metrics collected • User feedback analysed	• Testing & Evaluation Report • Performance Benchmark Results • UAT Summary
Final Documentation & Submission	15–16	• Final system refinement • Research documentation and presentation	• Final system approved • Research submission ready	• Final Research Report • SafeSight Prototype • Future Recommendations

6. Implemented Features: Full-Stack Workplace Safety Ecosystem

SafeSight is architected and implemented a comprehensive, multi-layered workplace safety platform that seamlessly transitions from a public-facing landing page to a secure, data-driven analytics engine. Every component from front-end user interface design to backend SQL architecture was purpose-built to support the safety operations of the Fairmont Waterfront.

Landing Page & Information Architecture

- AIDA Framework Integration

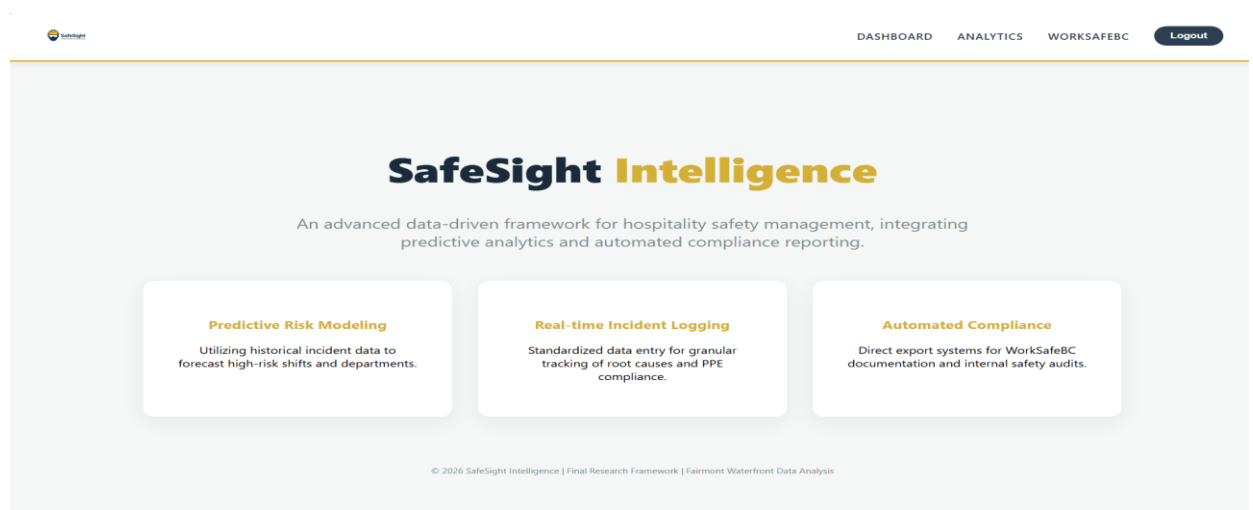
The homepage was strategically structured using the AIDA (Attention, Interest, Desire, Action) marketing framework to effectively engage stakeholders and communicate the value of real-time safety intelligence.

- Responsive React Architecture

Developed modular, reusable React components to ensure consistent branding, optimized performance, and full responsiveness across desktop, tablet, and mobile devices.

- Strategic Call-to-Action Design

Designed a streamlined navigation flow guiding authorized users directly to the secure login portal, minimizing friction while maintaining clarity of purpose.



Secure Authentication & Role-Based Access

- Node.js Authentication Gateway:

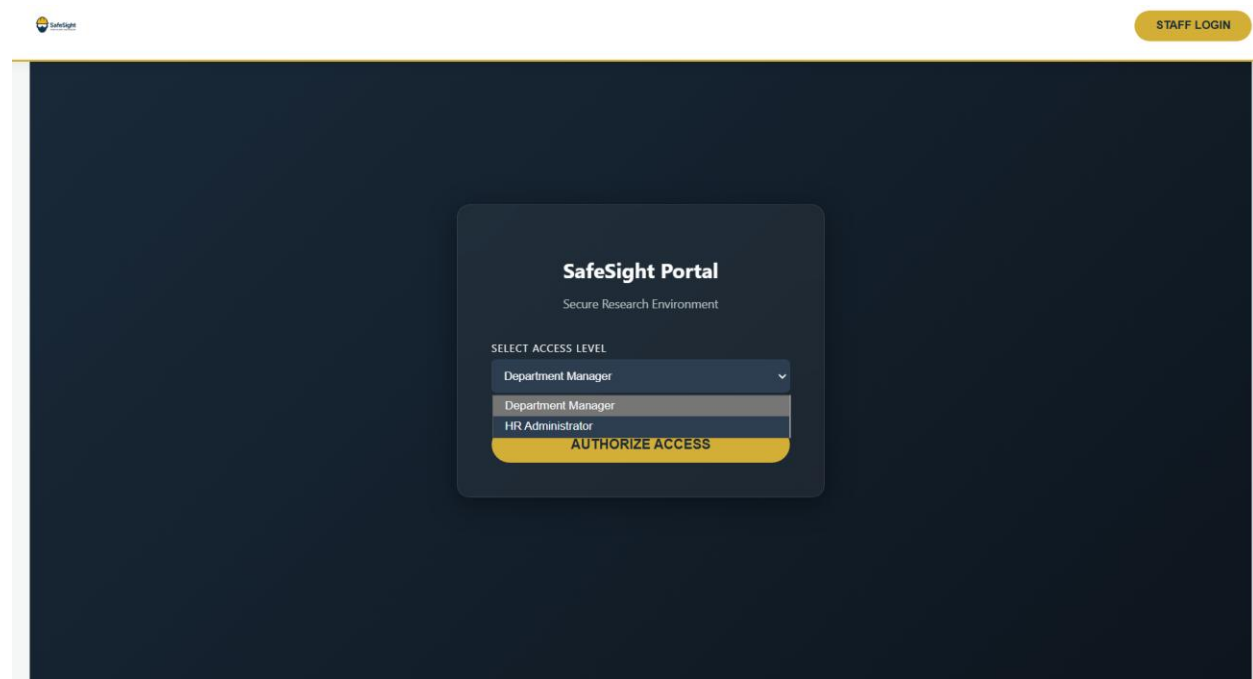
Implemented a secure authentication system powered by a Node.js server (Port 5000), enabling encrypted communication between client and server layers.

- Role-Based Access Configuration:

Established a defined “HR Research Access” permission level to enable high-level auditing and controlled access to historical safety data.

- Enterprise-Level Data Protection:

Designed access controls to ensure that 391 historical incident records and 345 hospitalization records are exclusively available to verified administrative users, reinforcing data confidentiality and compliance standards.



Advanced Analytics Dashboard

The analytics dashboard serves as the operational intelligence hub of the platform, integrating real-time data processing with executive-level visualization.

- Executive KPI Framework:

Engineered a dynamic “stat card” system highlighting three mission-critical metrics:

- Total Incidents: 391
- Critical Risk Cases: 380
- Hospitalization Cases: 345

- Severity Distribution Visualization (Recharts):

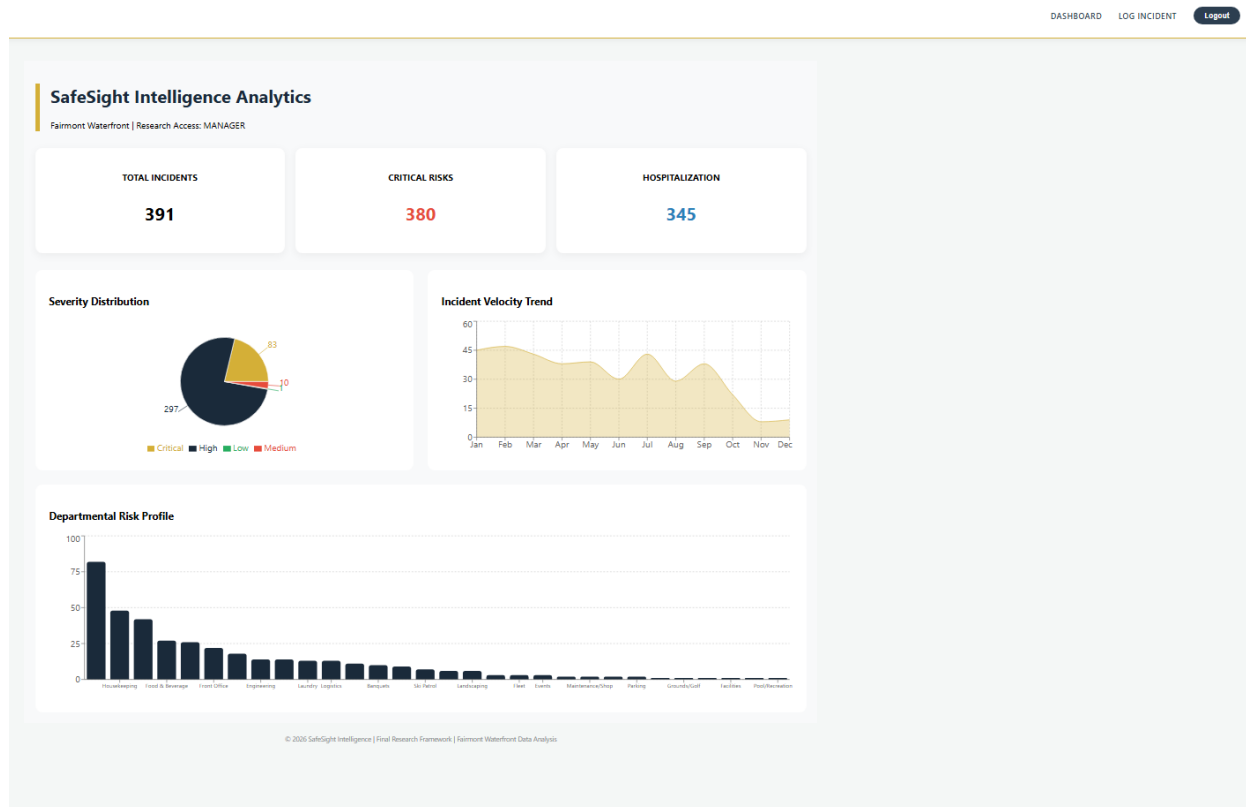
Developed an interactive pie chart using Recharts to display incident severity distribution, identifying 297 cases classified as “High” risk.

- Incident Velocity Analysis:

Implemented a time-series area chart to track monthly fluctuations in incident frequency, revealing peak activity in February and stabilization during summer months.

- Departmental Risk Profiling:

Designed a comparative bar chart ranking safety performance across 20+ departments, identifying Housekeeping and Food & Beverage as priority areas for risk mitigation initiatives.



Incident Intake & Database Management

- **Full-Stack Reporting Workflow:**

Built a custom React-based incident reporting form capable of capturing detailed operational data, including department classification and PPE compliance.

- **Relational Database Architecture (MySQL):**

Designed and structured the safesight_db schema to support complex relational mapping between incidents, departments, and hospitalization data.

- **Data Integrity & Type Conversion Logic:**

Implemented backend logic to convert frontend boolean inputs into SQL-compatible TINYINT values, ensuring consistency, integrity, and accurate data storage across the system.


DASHBOARD
LOG INCIDENT
Logout

Advanced Safety Incident Reporting

Department
Housekeeping

Specific Work Area
e.g. Lobby, Kitchen Line, Room 402

Shift
Morning

Incident Date & Time
yyyy-mm-dd --:--:--

Incident Type
Slip/Fall

Severity Level
Low

Detailed Description
Describe exactly what happened...

Initial Root Cause Assessment
e.g. Missing wet floor sign, Improper lifting technique

☐ Was PPE being used?
☐ Immediate Hospitalization?

LOG INCIDENT

© 2026 SafeSight Intelligence | Final Research Framework | Fairmont Waterfront Data Analysis

1.1.5 Technical Implementation Details

Layer	Technology	Individual Contribution
Frontend	React.js, Recharts	Designed the UI, implemented the AIDA framework, and built the data visualizations.
Backend	Node.js, Express	Developed the API architecture, handled CORS protocols, and managed server-side logic.
Database	MySQL	Architected the relational schema, performed data cleaning on Fairmont datasets, and optimized queries.

Code Implementation Samples

Backend API & Server Configuration (Node.js)

This backend snippet demonstrates the primary entry point of the application (server.js). It highlights the implementation of CORS middleware to facilitate cross-origin communication with the React frontend and the mounting of centralized analytics routes on Port 5000.

```

1  const express = require('express');
2  const cors = require('cors');
3  const analyticsRoutes = require('./routes/analyticsRoutes');
4
5  const app = express();
6  app.use(cors());
7  app.use(express.json());
8
9  // Mount the routes
10 app.use('/api', analyticsRoutes);
11
12 app.get('/', (req, res) => {
13   res.send('SafeSight Backend is running!');
14 });
15
16 const PORT = 5000;
17 app.listen(PORT, () => {
18   console.log(`🚀 Server started on port ${PORT}`);
19 });

```

Database Schema & Data Integrity (MySQL)

Detailed MySQL schema for the incidents table in the SafeSight database. The architecture includes critical data fields such as `incident_datetime`, `severity`, and a `tinyint(1)` flag for hospitalized status to ensure efficient storage and precise querying for the frontend analytics dashboard.

Result Grid						
Filter Rows:						
Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
►	incident_id	int	NO	PRI	NULL	auto_increment
	department	varchar(100)	YES		NULL	
	work_area	varchar(100)	YES		NULL	
	shift	varchar(50)	YES		NULL	
	incident_datetime	datetime	YES		NULL	
	incident_type	varchar(100)	YES		NULL	
	description	text	YES		NULL	
	severity	varchar(50)	YES		NULL	
	root_cause	text	YES		NULL	
	ppe_worn	varchar(100)	YES		NULL	
	hospitalized	tinyint(1)	YES		NULL	
	status	varchar(50)	YES		Repor...	
	created_at	timestamp	YES		CURR...	DEFAULT_GEN...
	incident_month	varchar(100)	YES		NULL	

Data Visualization Logic (React & Recharts)

React.js implementation of the Departmental Risk Profile component using the Recharts library. The code demonstrates the use of a `<ResponsiveContainer>` to ensure dashboard fluidity, mapping aggregated backend data to a customized `<BarChart>` with specific styling for data keys and axes.

```

{/* 3. Departmental Risk Profile (Bar Chart) */}
<div style={{ background: 'white', padding: '25px', borderRadius: '12px', gridColumn: 'span 2' }}>
  <h3>Departmental Risk Profile</h3>
  <ResponsiveContainer width="100%" height={300}>
    <BarChart data={data.departmentData || []}>
      <CartesianGrid strokeDasharray="3 3" vertical={false} />
      <XAxis dataKey="name" tick={{fontSize: 10}} />
      <YAxis />
      <Tooltip />
      <Bar dataKey="value" fill="#1a2a3a" radius={[4, 4, 0, 0]} />
    </BarChart>
  </ResponsiveContainer>
</div>

```

Frontend State Management & API Fetching (React)

React functional component utilizing the `useEffect` hook to perform an asynchronous fetch request to the Node.js API (Port 5000), retrieving 391 historical records for real-time dashboard state management.

```
11 useEffect(() => {  
12   fetch('http://localhost:5000/api/dashboard-stats')  
13     .then(res => res.json())  
14     .then(result => {  
15       setData(result);  
16       setLoading(false);  
17     })  
18     .catch(err => {  
19       console.error("Fetch Error:", err);  
20       setLoading(false);  
21     });  
22 }, []);
```

7. Acknowledgement

I would like to express my sincere gratitude to my instructor, Padmapriya Arasanipalai Kandhadai (Priya), for providing the initial idea for this project and for guiding me in refining the scope to ensure its academic and practical relevance.

I would also like to acknowledge Fairmont Waterfront for serving as the organizational context for this project. The operational environment of the hotel provided a realistic and relevant foundation for designing and evaluating the SafeSight workplace safety management platform.

(All organizational references used in this project are for academic purposes only.)

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Appendix 1: Work Log

Date	Number of Hours	Description	Remarks
January 9, 2026	3	Initial class of the CSIS 4495	Initial idea of the research project of the HRIS platform extension was extended to Prof. Priya and she gave the idea of Work Safety Dashboard
January 10, 2026	1	Research done on the worksafe practices and the availability of the similar dashboards	
January 12, 2026	0.3	Spoke to the HR of the Fairmont Waterfront to get the data	
January 15, 2026	0.5	searched about the technology that I have to use and created the Git hub Repo and sent the collaboration request to Prof. Priya	
January 16, 2026	3	Project discussion with Prof. Priya and Finalizing the scope	
January 20, 2026	2	Project Proposal draft	
January 21, 2026	1	Project Proposal draft	
January 22, 2026	0.5	Project Proposal draft	
January 24, 2026	2	Project Proposal draft	
January 25, 2026	2	Finalizing the research project	
January 30, 2026	2	Mandatory consultation with supervisor	Discussed project direction and received guidance to obtain operational requirements from Fairmont Waterfront

February 1, 2026	1	Development environment setup	Configured local development environment and required tools for system development
February 2, 2026	2	Meeting with Fairmont Waterfront HR department	Collected operational insights from HR to prioritise system features, as advised by the supervisor
February 3, 2026	2	Developed Backend (Application Layer) using Node.js and Express.	Installed dependencies and wrote index.js for DB connectivity.
February 4, 2026	1	Dataset and industry research	Conducted in-depth research on relevant datasets and reviewed comparable workplace safety systems
February 5, 2026	2	Dashboard feature research	Analysed and shortlisted core features required for the SafeSight dashboard
February 6, 2026	2	Dataset cleaning and SQL import	Cleaned selected datasets and prepared them for import into the SQL database
February 7, 2026	1	Integrated React.js Frontend (Presentation Layer).	Scaffolded the React app inside the FrontEnd directory.
February 7, 2026	0.5	Verified Full-Stack connection and committed changes to GitHub. Confirmed "Connected to SafeSight MySQL Database" in terminal.	
February 8, 2026	1	SafeSight data preparation (Jupyter Notebook)	Developed and documented data preparation steps using a Jupyter Notebook
February 9, 2026	3	Progress Report 1 finalisation	Compiled, reviewed, and finalised Progress Report 1 for submission
February 9, 2026	3	Phase 2 headstart with bridging presentation layer and the data layer	

February 10, 2026	2	Database Schema Optimization	Refined MySQL tables to ensure data integrity for incident severity mapping.
February 11, 2026	1.5	API Endpoint Development	Created Node.js routes to fetch total incident counts and hospitalization data.
February 12, 2026	2	Frontend Dashboard Integration	Integrated Recharts library into React to display the Incident Velocity Trend.
February 13, 2026	1.5	Component Styling	Designed the "SafeSight Intelligence Analytics" header and high-level stat cards.
February 14, 2026	2.5	Data Mapping: Fairmont Waterfront	Successfully mapped 391 historical records to the dashboard visualization layer.
February 15, 2026	2	Severity Distribution Logic	Programmed the pie chart logic to categorize "Critical" (83) and "High" (297) risks.
February 16, 2026	2	Departmental Risk Profiling	Developed the bar chart component to identify Housekeeping as the top risk area.
February 17, 2026	1.5	Research Access Configuration	Implemented "HR" research access level labels and restricted views in the UI.
February 18, 2026	2	Midterm Presentation Planning	Outlined the 10-minute presentation script for the full-stack system.
February 19, 2026	1	Navigation & UI Logic	Finalized the top navigation bar links for Dashboard, Analytics, and WorkSafeBC.

February 20, 2026	2.5	Analytics Calculations	Verified the 97.1% risk probability metric against the 380 identified critical risks.
February 21, 2026	2	Script Refinement	Developed detailed speaking notes for role-based features and technical architecture.
February 22, 2026	2	Roadmap & Worklog Finalization	Documented future AI implementation plans and finalized the Phase 2 development log.

Appendix 2: AI Use Section

AI Name	Tool	Version / Account	Specific Feature Used	Prompt (Summary)	Used Value Addition (Student Contribution)
ChatGPT		GPT-5.2 / Free	Idea generation & academic structuring	“Suggest HRIS add-ons related to workplace safety systems”	Evaluated relevance, selected only safety-focused HRIS features, and adapted them to Fairmont Waterfront context
ChatGPT		GPT-5.2 / Free	Comparative analysis	“List available work and safety systems used in hospitality”	Analysed limitations of existing systems and identified gaps SafeSight addresses
ChatGPT		GPT-5.2 / Free	Academic writing support	“Provide a format for a literature review”	Reorganised structure to meet postgraduate marking rubric and project scope
ChatGPT		GPT-5.2 / Free	Analytical framework explanation	“Explain types of gap analysis for information systems”	Selected operational and compliance gap analysis suitable for WorkSafeBC context
ChatGPT		GPT-5.2 / Free	Research methodology structuring	“Provide format of a research design”	Tailored design to technical system development rather than theoretical research
ChatGPT		GPT-5.2 / Free	Methodology guidance	“How to form a research design for technical development research”	Aligned SDLC phases with academic research expectations
ChatGPT		GPT-5.2 / Free	Research methods suggestion	“Data collection methods for technical research projects”	Selected simulated datasets and system-generated data relevant to safety analytics
ChatGPT		GPT-5.2 / Free	Research validation strategy	“Explain triangulation strategies in applied research”	Applied triangulation using system output, predictive results, and usability feedback

ChatGPT	GPT-5.2 / Free	Technology recommendation	“Identify technology stack for a full-stack predictive dashboard”	Finalised stack based on feasibility, course scope, and personal skill level
ChatGPT	GPT-5.2 / Free	Code generation	“Generate Mermaid code for Kanban timeline”	Validated task logic, adjusted timelines, and aligned with Gantt and PERT plans
ChatGPT	GPT-5.2 / Free	System design explanation	“Describe three-tier system architecture for a safety system”	Adapted architecture specifically for HR, manager, and reporting portals
ChatGPT	GPT-5.2 / Free	Formal document drafting	“Draft a project contract for academic proposal”	Simplified language, aligned with course outline, and customised for instructor
M365 Copilot	GPT-5 chat model / Free	Logo creation	"create a logo for work safety dashboard name: SafeSight"	
ChatGPT	GPT-5.2 / Free	Database design guidance	“Suggest database structure for a workplace safety analytics system”	Reviewed proposed schema, normalised tables, and selected entities relevant to hospitality safety incidents
ChatGPT	GPT-5.2 / Free	Data modelling explanation	“Explain relational vs analytical database models”	Chose relational model for transactional safety data and justified design choice in report
ChatGPT	GPT-5.2 / Free	Dataset cleaning guidance	“How to clean and prepare incident datasets for predictive analysis”	Applied only applicable cleaning steps and excluded unrealistic transformations for simulated data
ChatGPT	GPT-5.2 / Free	Data validation rules	“What validation checks are required for safety datasets”	Defined validation rules aligned with workplace incident reporting standards
ChatGPT	GPT-5.2 / Free	Front-end architecture explanation	“Explain front-end architecture for a data dashboard”	Selected dashboard-based UI approach suitable for managers and HR users

ChatGPT	GPT-5.2 / Free	Back-end design support	“Describe back-end services for analytics systems”	Mapped suggested services to a simplified API structure appropriate for course scope
ChatGPT	GPT-5.2 / Free	API design guidance	“Design REST API endpoints for a safety reporting system”	Filtered endpoints to core MVP features and documented them at a conceptual level
ChatGPT	GPT-5.2 / Free	Environment setup guidance	“Development environment setup for full-stack analytics project”	Selected tools and configurations compatible with local machine and institutional constraints
ChatGPT	GPT-5.2 / Free	Deployment awareness	“Common deployment considerations for academic system projects”	Identified constraints and documented limitations rather than attempting premature deployment
ChatGPT	GPT-5.2 / Free	Progress report structuring	“Provide a format for Progress Report 1 for a system development project”	Selected only sections required by course guidelines and adapted structure to reflect SafeSight’s early development stage
ChatGPT	GPT-5.2 / Free	Milestone planning	“Identify suitable milestones for Phase 1 of a predictive system project”	Refined milestones to align with completed research activities and realistic academic timelines
ChatGPT	GPT-5.2 / Free	Scope clarification	“Explain how to define project scope for an applied IT research project”	Clarified system boundaries and excluded advanced features not feasible within course duration
ChatGPT	GPT-5.2 / Free	Requirement prioritisation	“How to prioritise functional requirements for an MVP system”	Evaluated suggested requirements and selected core safety analytics features for initial implementation
ChatGPT	GPT-5.2 / Free	Risk and issue identification	“Common early-stage risks in system development projects”	Identified relevant risks and documented mitigation strategies

				applicable to SafeSight development
ChatGPT	GPT-5.2 / Free	Academic reflection support	"How to write reflective progress for an IT project"	Rewrote content in own words, ensuring critical reflection rather than descriptive reporting
ChatGPT	GPT-5.2 / Free	Alignment check	"How to align progress reports with original project proposal"	Cross-checked progress against proposal objectives and adjusted report to maintain consistency
Gemini	Paid	Technical Troubleshooting	"Troubleshooting Node"	Resolved backend connectivity issues and port 5000 configuration.
Gemini	Paid	Full-Stack Integration	"Bridging the layers"	Successfully linked MySQL queries to React frontend components.
Gemini	Paid	Data Visualization	"Dashboard stats and Recharts"	Developed logic for Severity Distribution and Velocity Trend charts.
Gemini	Paid	UX Design Analysis	"AIDA framework for Dashboard"	Applied information hierarchy to prioritize Critical Risks (380) and Hospitalization (345) stats.
Gemini	Paid	Business Logic	"Departmental Risk Profile"	Identified Housekeeping as the primary risk area with 82+ incidents.
Gemini	Paid	Content Creation	"10-minute presentation script"	Structured a professional narrative for the solo midterm presentation.